



Yakeen Plus (2025)

Short Practice Test - 07

DURATION : 60 Minutes

DATE : 30/10/2024

M. MARKS : 192

ANSWER KEY

PHYSICS

1. (3)
2. (1)
3. (4)
4. (2)
5. (3)
6. (1)
7. (2)
8. (1)
9. (2)
10. (1)
11. (1)
12. (3)

CHEMISTRY

13. (2)
14. (2)
15. (1)
16. (1)
17. (4)
18. (2)
19. (4)
20. (3)
21. (2)
22. (3)
23. (2)
24. (1)

BOTANY

25. (4)
26. (3)
27. (3)
28. (3)
29. (1)
30. (3)
31. (3)
32. (1)
33. (1)
34. (4)
35. (4)
36. (3)

ZOOLOGY

37. (3)
38. (1)
39. (2)
40. (3)
41. (4)
42. (4)
43. (1)
44. (3)
45. (3)
46. (1)
47. (2)
48. (1)

SECTION-(I) PHYSICS

1. (3)

$$J = \frac{i}{A}, \text{ Current remains same}$$

$$J \propto \frac{1}{A}$$

$$\frac{J_P}{J_Q} = \frac{A_Q}{A_P} = \frac{\pi r_Q^2}{\pi r_P^2}$$

$$\frac{J_P}{J_Q} = \left(\frac{R}{2R}\right)^2 = \frac{1}{4}$$

[New NCERT Class 12th Page No. 99]

2. (1)

$$P = \frac{V^2}{R}$$

$$P \propto \frac{1}{R}$$

$$\frac{P_1}{P_2} = \frac{R_2}{R_1}$$

$$\boxed{\frac{P_1}{P_2} = \frac{2}{1}}$$

[New NCERT Class 12th Page No. 110]

3. (4)

$$V_A - 2 \times 2 - 3 - 2 = V_B$$

$$\boxed{V_A - V_B = 9V}$$

[New NCERT Class 12th Page No. 113]

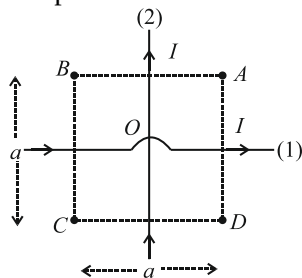
4. (2)

$$\int \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{inside}} = \mu_0 (4i - i) = 3\mu_0 i$$

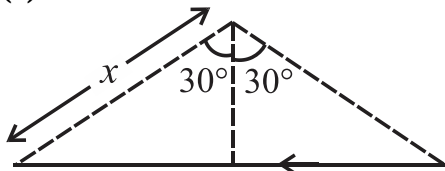
[New NCERT Class 12th Page No. 147]

5. (3)

Direction of magnetic fields at point A and C is opposite but having same magnitude so magnetic field at these points is zero.

[New NCERT Class 12th Page No. 119]

6. (1)



$$B = \frac{\mu_0 I}{4\pi x \cos 30^\circ} [\sin 30^\circ + \sin 30^\circ]$$

$$B = \frac{\mu_0 I}{4\pi x \left(\frac{\sqrt{3}}{2}\right)} \left(\frac{1}{2} + \frac{1}{2}\right) = \frac{\mu_0 I}{2\sqrt{3}\pi x}$$

[New NCERT Class 12th Page No. 144]

7. (2)

$$\phi = 6t^2 - 5t + 1$$

$$\varepsilon = -\frac{d\phi}{dt} = -12t + 5$$

$$I = \frac{\varepsilon}{R} = \frac{-12\left(\frac{1}{4}\right) + 5}{10} = 0.2A$$

[New NCERT Class 12th Page No. 180]

8. (1)

Electric field is produced by time varying magnetic flux.

[New NCERT Class 12th Page No. 182]

9. (2)

From Faraday's second law, emf induced in the circuit

$$e = \frac{\Delta\phi}{\Delta t}$$

If R is the resistance of the circuit, then

$$i = \frac{e}{R} = \frac{\Delta\phi}{R\Delta t}$$

Thus, charge passing through the circuit,

$$q = i \times \Delta t$$

$$\Rightarrow q = \frac{\Delta\phi}{R\Delta t} \cdot \Delta t$$

$$\Rightarrow q = \frac{\Delta\phi}{R}$$

[New NCERT Class 12th Page No. 190]

10. (1)

Magnetic moment = pole strength $\times$ length

[New NCERT Class 12th Page No. 137]

11. (1)

$$\chi = \frac{c}{T}, \chi \text{ is susceptibility}$$

$$\chi = m \left(\frac{1}{T}\right)$$

$$\tan \theta = m = \frac{0.4}{7 \times 10^{-3}} = 57 \text{ K}$$

[New NCERT Class 12th Page No. 148]

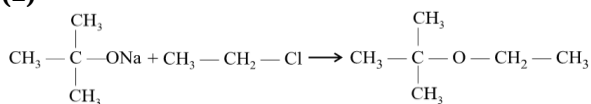
12. (3)

For diamagnetic substance $\chi_m < 0$ and $\mu_r < 1$

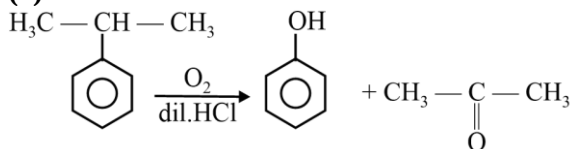
[New NCERT Class 12th Page No. 149]

SECTION-(II) CHEMISTRY

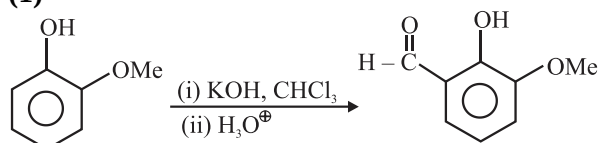
13. (2)

[New NCERT Class 12th Page No. 215]

14. (2)

[New NCERT Class 12th Page No. 202]

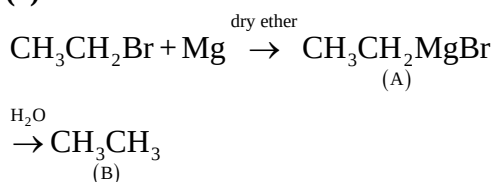
15. (1)



(Reimer – Tiemann reaction)

[New NCERT Class 12th Page No. 209]

16. (1)

[New NCERT Class 12th Page No. 181]

17. (4)

$\text{S}_{\text{N}}2$ reactions of optically active halides are accompanied by inversion of configuration.
 $\text{S}_{\text{N}}1$ reactions of optically active halides are characterized by racemisation.

[New NCERT Class 12th Page No. 175]

18. (2)

Aryl halides are less reactive towards nucleophilic substitution reactions than alkyl halides due to resonance stabilization.

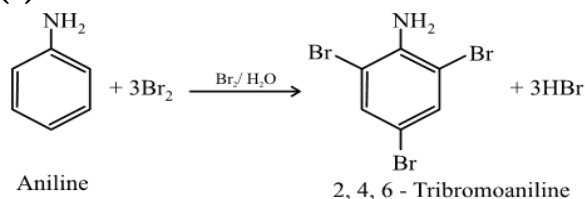
[New NCERT Class 12th Page No. 182]

19. (4)

The carbylamine test is given by primary amine in which it is converted into isocyanide.
 2, 4-dimethyl aniline and p-methyl benzylamine both are primary amines. Primary amines do give a positive carbylamine test.

[New NCERT Class 12th Page No. 271]

20. (3)

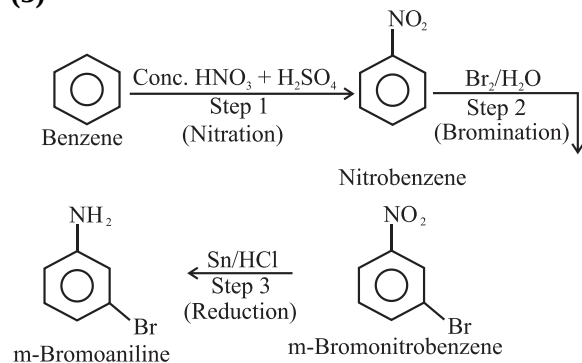
[New NCERT Class 12th Page No. 264]

21. (2)

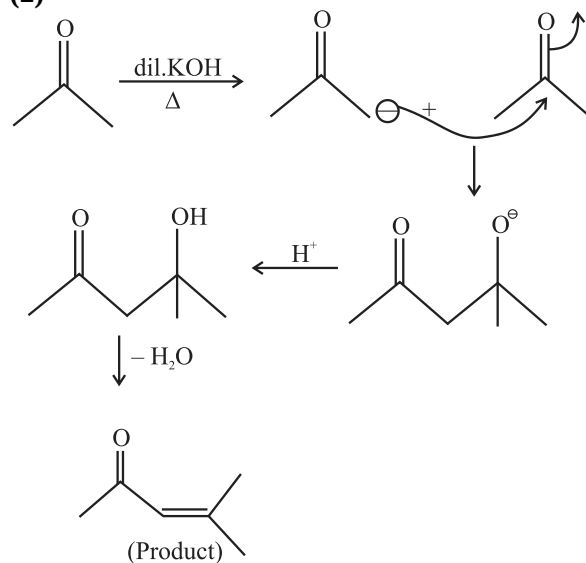
Nitrous acid reacts with hydrochloric acid (HCl) in the reaction mixture to generate the key intermediate, nitrosonium ion (NO^+). This electrophilic ion plays a crucial role in the diazotization process by attacking the aromatic ring of the amine and forming the diazonium salt.

[New NCERT Class 12th Page No. 273]

22. (3)

[New NCERT Class 12th Page No. 273]

23. (2)

[New NCERT Class 12th Page No. 241]

24. (1)

LiAlH_4 will reduce $-\text{CHO}$ into $-\text{CH}_2\text{OH}$ but it will not reduce the conjugated double bond.

[New NCERT Class 12th Page No. 238]

SECTION-(III) BOTANY

25. (4)
- | | |
|----------|-------------------|
| Ethylene | Gases(C_2H_4) |
|----------|-------------------|
- [New NCERT Class 11th Page No. 174]

26. (3)
- More than 100 gibberellins reported from widely different organisms such as fungi and higher plants.
- [New NCERT Class 11th Page No. 176]

27. (3)
- Formation of NADPH is associated with photosystem-I.
- [New NCERT Class 11th Page No. 139]

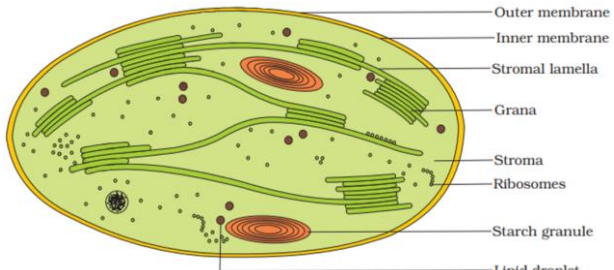
28. (3)
- Most zygotes divide only after certain amount of endosperm is formed.
 - Endosperm formation is an adaptation to provide assured nutrition to the developing embryo.
- [New NCERT Class 12th Page No. 19]

29. (1)
- In arithmetic growth rate, when length of the organ is plotted against time, the nature of graph curve will be linear.
- [New NCERT Class 11th Page No. 170]

30. (3)
- Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called plasticity, e.g., heterophylly in cotton, coriander and larkspur.
- [New NCERT Class 11th Page No. 173]

31. (3)
- The microsporangia is surrounded by four wall layers.
 - Each lobe of anther is dithecalous.
- [New NCERT Class 12th Page No. 5]

32. (1)
- The hard outer layer called the exine is made up of Sporopollenin. The exine exhibits a fascinating array of patterns and designs.
- [New NCERT Class 12th Page No. 8]

33. (1)
- 
- Trapping the light energy-Grana (II) and Stroma lamella (I).
 - Utilisation of ATP-Stroma (III).
- [New NCERT Class 11th Page No. 136]

34. (4)
- | | |
|-----------------------|---|
| <i>Vallisneria</i> | Female flowers reach the surface of water by the long stalk |
| Seagrasses | Female flowers remain submerged |
| Corn | Wind pollination |
| <i>Amorphophallus</i> | Provide safe place to lay eggs for insect |
- [New NCERT Class 12th Page No. 13, 14]

35. (4)
- The C_4 plants show saturation at about $360 \mu L^{-1}$ while C_3 responds to increased CO_2 concentration and saturation is seen only beyond $450 \mu L^{-1}$.
- [New NCERT Class 12th Page No. 150]

36. (3)
- Calvin cycle requires ribulose biphosphate, water and CO_2 during carboxylation.
- [New NCERT Class 12th Page No. 144]

SECTION-(IV) ZOOLOGY

37. (3)
- After ovulation, remaining part of the Graafian follicle is converted into corpus luteum.
 - The primary oocyte within tertiary follicle completes its first meiotic division.
 - The secondary oocyte forms a new membrane called zona pellucida surrounding it.
 - Sperm formation continues in old men.

- During pregnancy, all events of the menstrual cycle stop and there is no menstruation.
- [New NCERT Class 12th Page No. 26, 32, 33, 35, 36]

38. (1)
- | | |
|--------------|---|
| Parturition | Delivery of baby |
| Implantation | Attachment of blastocyst to endometrium |
| Menopause | Cessation of menstrual cycle |
| Birth canal | Cervical canal along with vagina |
- [New NCERT Class 12th Page No. 29, 35, 37, 38]

39. (2)
The correct sequence is;
I. Secretion of FSH
III. Growth of follicles
V. Sudden increase in level of LH
IV. Ovulation
II. Growth of corpus luteum
I → III → V → IV → II

[New NCERT Class 12th Page No. 33, 34, 35]

40. (3)
Prostate gland is an unpaired accessory gland of the male reproductive system.

[New NCERT Class 12th Page No. 28]

41. (4)
- Rate of the chemical reaction doubles on increasing the temperature by 10°C.
 - Catalysed reactions proceed at rates higher than that of uncatalysed reactions.

[New NCERT Class 11th Page No. 113]

42. (4)
- Almost all enzymes are proteins.
 - There are some nucleic acids that behave like enzymes.
 - These are called ribozymes.

[New NCERT Class 11th Page No. 112]

43. (1)
- Exoskeletons of arthropods, for example, have a complex polysaccharide called chitin. These complex polysaccharides are mostly homopolymers.
 - Cellulose does not contain complex helices and hence cannot hold I₂.

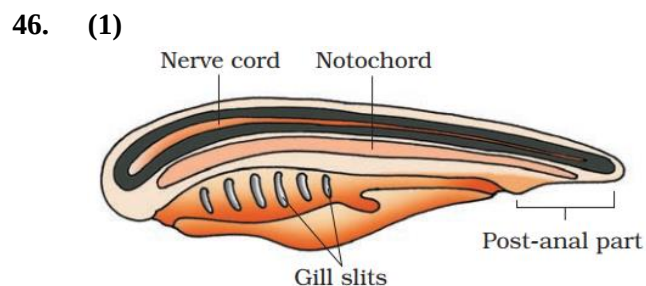
[New NCERT Class 11th Page No. 111, 110]

44. (3)
Ricin is a toxin.

[New NCERT Class 11th Page No. 108]

45. (3)
Wuchereria: Filarial worm
Ancylostoma: Hookworm

[New NCERT Class 11th Page No. 43]



[New NCERT Class 11th Page No. 46]

47. (2)
In cephalochordates, notochord extends from head to tail.

[New NCERT Class 11th Page No. 46]

48. (1)
- Class- Osteichthyes includes both marine and fresh water fishes with bony endoskeleton.
 - Air bladder is present which regulates buoyancy.
 - Examples: Marine – *Exocoetus* (Flying fish), *Hippocampus* (Sea horse); Freshwater – *Labeo* (Rohu), *Catla* (Katla), *Clarias* (Magur); Aquarium – *Betta* (Fighting fish), *Pterophyllum* (Angel fish).

[New NCERT Class 11th Page No. 48]

